2025 1st International Conference on Secure IoT, Assured and Trusted Computing (SATC) | 979-8-3315-1420-4/25/$31.00 ©2025 IEEE | DOI: 10.1109/SATC65530.2025.11137084

A Comprehensive Survey of Model Extraction Attacks: Current Trends, Defenses, and Future Directions

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***Abstract*—Model extraction attacks pose a significant threat to Machine Learning (ML) systems, especially in cloud-based services like Machine Learning as a Service (MLaaS). Attacks aim to steal proprietary models by replicating their functionality or extracting their internal parameters. This paper reviews model extraction attack types, examining existing defensive techniques and weaknesses in current defenses. Promising defense mechanisms are discussed, including adaptive privacy budgets, hybrid defense strategies, combining multiple methods, and hardware-based security solutions. Emerging attack models like collaborative attacks in federated learning environments are explored. Future research focuses on adaptive defenses and Artificial Intelligence (AI)-driven detection methods to improve model robustness and contribute to more resilient machine learning systems.**

***Index Terms*—Model extraction attacks, MLaaS security, adversarial machine learning, federated learning security.**

I. Introduction

An adversary attempting to replicate a machine learning (ML) model by repeatedly querying it and observing the results is known as a model extraction attack [1]. The main aim is to steal a model's functionality by trying to understand its decision boundaries. These attacks are very hazardous, especially in black-box environments where the adversary needs to know only about the model's input and output behavior. If these queries are analyzed and crafted for extraction attempts, the attackers have the upper hand. In this case, the security is breached by revealing sensitive intellectual property [2], [3]. Model extraction attacks pose several risks, which include:

- Intellectual Property Theft: ML models often involve data and computational resources investments. Successful model extraction attacks allow attackers to replicate these models, reducing the owner's competitive advantage [4].
- Security Risks: To bypass defenses, adversaries can generate adversarial examples that rely on these models [5].
- Privacy Concerns: Model extraction attacks and model inversion attacks can be combined to recover information about training data [4], [5].

A. Motivation

Using Machine Learning as a Service (MLaaS) platforms has led to increased accessibility to ML models but has also introduced new vectors for attacks, especially model extraction attacks. Understanding the trends in attack methodologies and defenses is critical to improving the security and privacy of machine learning systems. Current defenses, for instance, perturbation and rate-limiting, offer protection but have limitations in real-world scenarios. The trade-off between model security and model utility is a key challenge that must be solved. Therefore, evaluating existing protection mechanisms and suggesting hybrid and adaptive defense solutions that can keep up with the changing nature of model extraction attacks is imperative.

B. Gaps in Existing Research

The current defense techniques are either too computationally expensive for implementation or fail to address adaptive attacks where attackers change their strategies based on observing defenses. Moreover, current research only focuses on single-mode defenses, such as Differential Privacy (DP) alone, neglecting the potential for multi-layered systems that combine various approaches to fortify the overall security. Additionally, hybrid or side-channel extraction strategies have received less attention than query-based extraction [6], [7].

C. Contributions of This Survey Paper

This survey paper provides a comprehensive analysis of model extraction attacks, focusing on their types, techniques,

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and the vulnerabilities they exploit in MLaaS and other Artificial Intelligence (AI) systems. The attacks are categorized into query-based, side-channel, and data-free extraction attacks [8]. Current defense mechanisms, such as monitoring-based defenses [9], perturbation techniques are compared in detail, and their advantages and disadvantages are discussed.

In addition to reviewing the state-of-the-art defenses, this paper suggests novel defense mechanisms that comprise adaptive privacy budgets, hybrid defense systems, and hardware-assisted security measures to enhance protection against model extraction attacks. Furthermore, new attack forms, such as hybrid and multi-adversary attacks, are explored in federated learning systems [8] and recommendations for future research directions are offered in this paper. This survey intends to direct the creation of more reliable and safe machine learning systems by addressing the gaps in existing research and offering creative alternatives.

II. Related Work

In the field of machine learning security, model extraction attacks have become a crucial topic, particularly with the rise of MLaaS platforms. Several studies have explored different factors of model extraction attacks and their respective defenses.

A. Overview of Model Extraction Attacks

Early model extraction research concentrated on query-based attacks, in which adversaries try to replicate a machine learning model by repeatedly questioning it and determining its decision boundary. Black-box models can be replicated with high accuracy by simply using a large number of queries as demonstrated in Knockoff Nets proposed by Orekondy *et al.* in [10]. Jacobian-based augmentation technique is another query-based method that allows an adversary to systematically improve their model with minimal query overhead [11].

Another form of model extraction is a side-channel attack alongside query-based methods as detailed in the work of Chabanne *et al.* [12] where side-channel information such as power consumption was used to extract neural network architectures. Similarly, side-channel fuzzy analysis was used to exploit side-channel leaks in Internet of Things (IoT)-based models, offering another scope for attackers to reverse-engineer machine learning models.

B. Model Extraction in Federated Learning and Distributed Systems

Model extraction is a relatively new but growing field in the context of federated learning. Due to the constant updating of the model with data from numerous participants, federated learning systems pose special vulnerabilities [13]. The possibility that adversaries could use distributed model updates to extract models without having access to the original training data has been examined in "MEGEX: Data-Free Model Extraction" [8].

C. Defensive Mechanisms

In reply to these threats, various defense mechanisms have been proposed, each with flaws. Differential privacy has been widely researched as a mechanism to mitigate model extraction where noise is introduced to model outputs to prevent exact duplication. It is impractical for some applications as this method often leads to significant utility loss [14]. Monitoring-based techniques like Protecting Against DNN Model Stealing Attacks (PRADA) [15] detect suspicious question patterns but are susceptible to false positives. Hybrid defenses combining monitoring and differential privacy have been developed, but further study is needed.

III. Background and Preliminaries

A cloud-based platform called MLaaS provides sophisticated machine learning models and tools, such as image recognition and natural language processing, without requiring a lot of computing power. These platforms, which are offered by companies like Google, Amazon, and Microsoft, provide consumers black-box access so they may submit data and get predictions without being aware of the inner workings of the model [16], [17].

A. Attack Models in Model Extraction

Model extraction attacks can be classified into black-box and white-box attacks. Black-box attacks allow attackers to access input-output behavior through an Application Programming Interface (API), allowing them to duplicate the target model's functionality. White-box attacks provide full access to parameters and training data, often involving direct model interference. Without proper defense, MLaaS services can pose significant security risks [15], [18].

B. Query-Based and Side-Channel Attacks

Black-box environments often face query-based attacks, where adversaries create surrogate models by examining internal parameters. MEGEX, a data-free model extraction method, enhances this process [8]. Side-channel attacks exploit indirect information, like time or power consumption, to gain insight into model structure, especially in hardware-constrained scenarios like IoT devices, and successfully retrieve neural network topologies in edge computing applications [16]. Side-channel attacks can be combined with query-based attacks to bypass defense systems more easily and replicate ML models [15].

C. Typical Machine Learning Models Targeted

Model extraction attacks target Deep Neural Networks (DNNs), crucial in high-stakes applications like natural language processing, autonomous driving, and medical diagnostics, as they require significant processing power [18].

Apart from DNN, similar attacks can also target simpler machine learning methods like logistic regression, decision trees, and Support Vector Machines (SVM), which are less computationally demanding but used in corporate and research environments [15], [16]. Fig. 1 provides an overview of the model extraction process, illustrating how attackers can use query-based techniques to replicate proprietary machine learning models. The diagram illustrates the objectives and

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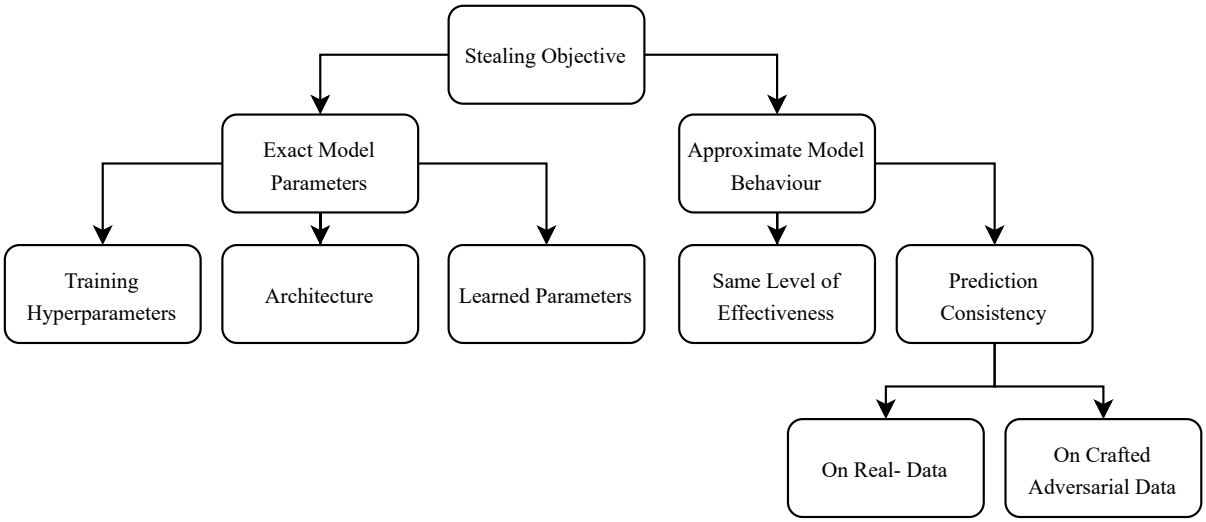


Fig. 1. Model extraction diagram.

methodologies of model extraction attacks, broadly categorized into two primary goals: exact model parameters and approximate model behavior [4]. Model extraction attacks are divided into training hyper-parameters, architecture, and learned parameters. Training hyper-parameters involves inferring hyper-parameters, while architecture stealing attacks target model structure. Learned parameters retrieve weights, affecting model secrecy. Approximate model behavior aims to outperform the target model on real data, while prediction consistency involves replicating the target model's predictions in hostile environments.

IV. Types of Model Extraction Attacks

Model extraction attacks can be categorized into Black-Box and White-Box attacks, with many branches, including Query-based, Side-channel, and Data-free. Black-box attacks rely on indirect methods or input-output observations. In contrast, white-box attacks can be adapted to certain techniques that require access to gradients in a white-box setting. These attacks can operate in various settings. Fig. 2 shows the overall categorization of model extraction attacks.

A. Query-Based Model Extraction Attacks

Model extraction attacks target to steal a model's internal parameters to steal the functionality of a model. The model's output behavior is targeted in functional extraction which includes decision boundaries and predictive accuracy, not necessarily duplicating the internal structure. When attackers are interested in creating a surrogate model that behaves identically to the target on given inputs known as query-based attacks [15], [19].

A primary example of query-based model extraction is **Knockoff Nets** [10]. An attacker can create a local model that resembles the target model's functionality using numerous input-output pairs obtained through repeated API queries. This method is effective in black-box settings and image classification applications, where high-fidelity replication is possible, as research shows Knockoff Nets can achieve near-parity [20]. **Jacobian-based attacks** maximize disparities between target and surrogate models by creating input samples that improve the surrogate model. By allowing sensible sampling, they optimize query efficiency, reduce API calls, and increase model quality in constrained query budgets [8].

B. Side-channel Attacks

Side-channel attacks can be further categorized into **Power Side-channel Attacks and Timing Side-channel Attacks.** Power side-channel attacks exploit power usage differences to understand machine-learning model operations during inference. They can extract architectural details like layers, parameters, and neuron activation patterns. These attacks are particularly effective in edge and IoT devices where models are placed near hardware. Power analysis exposes critical model structure aspects, compromising confidentiality and integrity, leading to security concerns [14], [18].

Timing side-channel attacks involve attackers analyzing model response times to infer structural or computational characteristics. Latency-infected models have deeper layers, revealing layer counts and configurations. These attacks can compromise models in black-box situations by inferring layer depths and activation functions, particularly in real-time applications where response time is crucial [19], [21].

C. Data-Free Model Extraction Attacks

Data-free model extraction attacks can be further classified into two types: **Gradient-based Technique and Synthetic Data Generation**. Gradient-based data-free attacks use synthetic data from gradient-based optimization techniques to approximate target models without real-world data. These attacks create high-fidelity surrogate models, optimizing information leaks. They are effective when attackers lack labeled data as they can determine model parameters and decision boundaries using model responses. Gradient-based methods accurately replicate deep learning models [20].

In machine learning, synthetic data generation is used to produce data that closely resembles actual data, particularly

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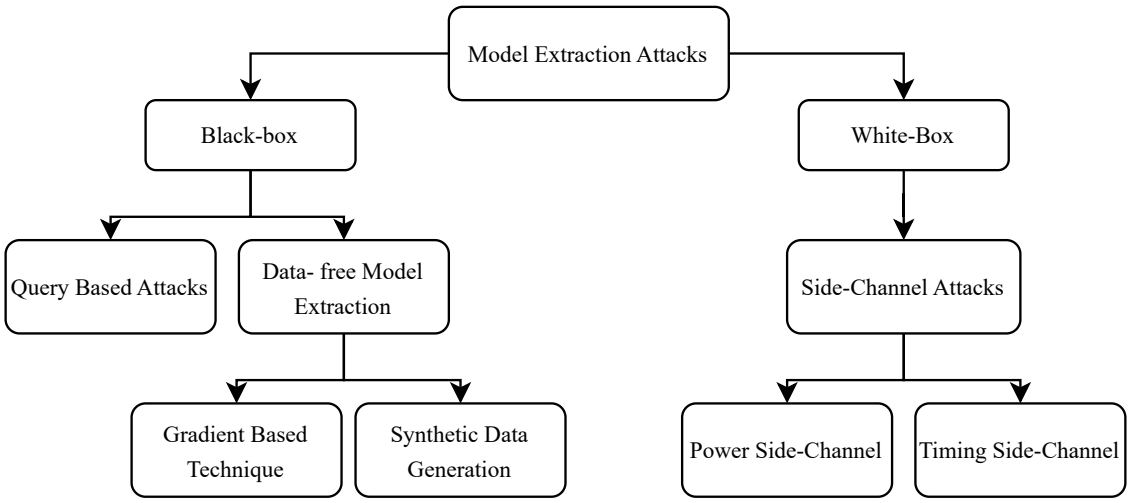


Fig. 2. Tree diagram of the categorization of model extraction attacks.

in fields like security and medical diagnostics where access to authentic datasets is limited, using techniques like Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs). Data augmentation techniques enable attackers to create artificial datasets mimicking a target model's training data, enhancing performance and broadening extraction in restricted searches [16], [17]. Gradient-based techniques rely on gradients of the target model's output to approximate decision boundaries and create synthetic inputs, while synthetic data generation generates artificial inputs using methods like random sampling and generative models without relying on gradient information.

V. Survey of Current Work

Model extraction attacks can be classified based on techniques including query-based, side-channel, and data-free methods. Survey studies by Oliynyk *et al.* [4] and Genç *et al.* [2] highlight these distinctions stating that each technique varies in its dependency on computational resources, data requirements, and query access. These techniques are very successful in real-life situations as shown in many studies. Knockoff Nets is a query-based attack in which adversaries mimic a target model's functionality using input-output pairs. Through repetitive API querying, this technique is successful in black-box settings within MLaaS systems [3], [22]. MEGEX improves the idea of data-free extraction by using generative models. In contrast to Knockoff Nets, MEGEX uses gradient-based information to approximate decision boundaries rather than labeled data. This method demonstrates that high-accuracy extraction is achievable in the absence of actual training data [23].

PRADA demonstrates the dual nature of query-based monitoring techniques by acting as both an attack and defense mechanism. As a defense, PRADA acts as a warning system by identifying question behaviors that are suspicious of model extraction attempts while as an attack, it demonstrates how attackers can carefully manipulate query patterns to evade fundamental monitoring protections [4], [21].

A. Current Defenses

The tree diagram in Fig. 3 shows the categorization of defense strategies in this paper.

The security and integrity of machine learning models are seriously threatened by model extraction attacks, particularly in MLaaS scenarios, where external users often access the models. Numerous defense techniques have been created to reduce these risks, and they can be divided into categories: access control and response manipulation. The following paragraphs discuss more about these defense mechanisms and how they are further categorized.

1) Access Control Defenses: The first line of security is access control measures, which limit the attack surface that adversaries might exploit by limiting models that can be accessed and used.

- **Query Limitation and Rate Limiting**: Limiting the number of inquiries per user is the goal of query limitation and rate limitation. This tactic helps reduce the viability of model extraction by preventing an attacker from acquiring enough samples.
- **Authentication and Authorization**: Implementing robust authentication measures ensures that only certain users have access to the model API adds a layer to illegal model extraction [17].

2) Response Manipulation Defenses: The implementation of strong authentication processes ensures that only those users who are authorized to do so can use the model and the API related to the model.

- **Output Perturbation**: The addition of noise or the alteration of the precision level of model output such as rounding the confidence values can tremendously reduce the accuracy of a reconstructed model. This method has been looked into by many researchers as a viable means of coping with extraction attempts with high accuracy, for instance, differential privacy [21], [24].
- **Randomized Output**: Attackers will find it more difficult to consistently retrieve the model's parameters if the

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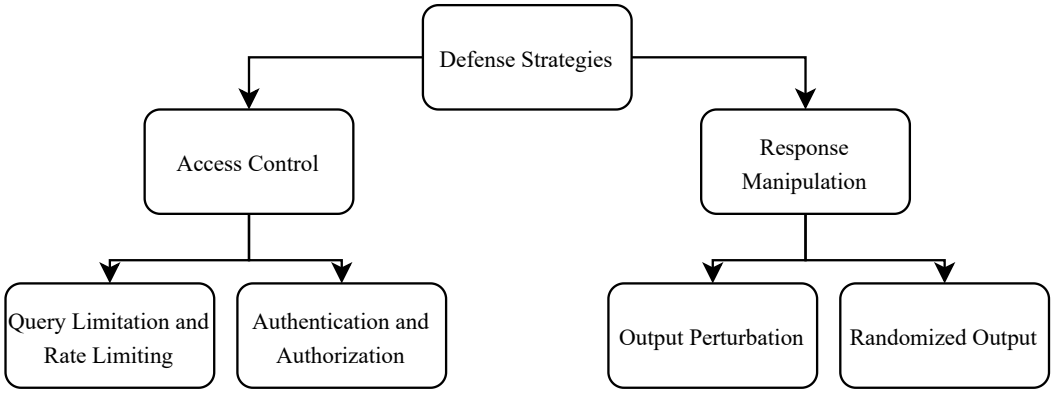


Fig. 3. Tree diagram of the categorization of defense mechanisms.

TABLE I
TYPES OF MODEL EXTRACTION ATTACKS AND CORRESPONDING DEFENSE MECHANISMS

Attack Type	Defense Mechanisms
Query-Based Attacks	Rate Limiting, Output Perturbation
Side-Channel Attacks	Randomized Processing Sequences
Data-Free Model Extraction	Query Pattern Analysis, Dynamic Differential Privacy
Hybrid Query	Hybrid Defenses
Collaborative Attacks	Federated Learning-Specific Defenses
Adaptive Attacks	Adaptive Privacy Budget Allocation

outputs are changed randomly. Although this approach is known to introduce a trade-off between security and usability, it works well in many situations [14].

Table I provides a comprehensive overview of attack types and their corresponding defenses, aiding in understanding model extraction threats and identifying suitable protection mechanisms.

VI. Limitations of Current Defenses

Each of the defense types has notable limitations. As model extraction techniques evolve, the attackers find unique ways to come up against these defenses. This section discusses the limitations in detail, shedding light on challenges in applying differential privacy effectively, the difficulties monitoring-based defenses have with adaptive adversaries, and the particular vulnerabilities of perturbation-based methods in restricting extraction under repeated queries. This overview presents the growing need for adaptable, hybrid solutions that convey each defense's shortcomings in isolation.

A. Inefficiency of Differential Privacy

Noise in model protection is a challenge, with balancing privacy budgets being a major challenge. Too much noise can degrade model utility, while too little weaken protection against model extraction, especially in fields like healthcare and finance [22]. Adaptive attacks, which use high-volume, dynamic querying, can bypass DP's safeguards by averaging outputs from multiple queries, exposing a vulnerability that DP excels in static environments but faces in dynamic scenarios [2], [4].

B. Limitations of Monitoring-Based Approaches

Monitoring-based defenses such as PRADA and differential privacy effectively detect model extraction, but face issues with false positives and negatives, and adaptive attackers can evade detection by slowing down queries [3]. Monitoring-based defenses face challenges from query flooding and evasion techniques, highlighting the need for flexible monitoring techniques. The shortcomings of PRADA highlight the need to distinguish between suspicious and typical activity, while differential privacy addresses adaptive attacks.

VII. Proposed New Defense Mechanisms

Researchers have developed innovative defense mechanisms for machine learning models, including hardware-supported, adaptive, and hybrid security solutions, ensuring models are protected in protected environments. Adaptive privacy budget allocation is a sophisticated perturbation technique that dynamically adjusts noise to maintain model functionality for authorized users, potentially safeguarding machine learning models against extraction attempts [4], [22].

1) Dynamic Noise Injection Based on Model Extraction Status: Dynamic noise injection is an extension of adaptive budgeting that adjusts the noise amount based on model extraction attempts. Examining query behavior for unusual frequency, the system assesses possible risks and adjusts noise appropriately. A more responsive alternative to static DP protections, dynamic noise injection obstructs extraction during continuous evaluation without significantly impacting legitimate queries [17], [18].

A. Hybrid Defense Systems

Hybrid defensive systems use differential privacy and monitoring-based defenses such as PRADA to handle access patterns and response integrity, effectively addressing the drawbacks of monitoring and DP in MLaaS contexts [4], [15].

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Enhanced anomaly detection combined with noise insertion, hybrid systems utilize machine learning-based anomaly detection to identify potential threats based on query behavior trends. This optimizes security and performance by selectively introducing noise when suspicious patterns are detected, increasing detection accuracy, and reducing false positives [2], [19].

B. Hardware-Assisted Security Measures

Trusted Execution Environments (TEEs) protect AI models by separating computations from physical devices, reducing risks of model extraction. However, special side-channel mitigation measures randomizing processing steps and power masking are needed to protect against indirect information attacks [25].

VIII. Future Work and Research Directions

The research community is focusing on developing robust defense mechanisms for model extraction techniques, aiming to strengthen defenses in decentralized and adaptable machine-learning environments. Key research topics include adaptive, multi-layered defenses that incorporate monitoring, perturbation, and differential privacy, as model extraction attacks become more complex. These defenses provide stronger defenses against high-volume, adaptable attacks by dynamically adapting to threat patterns [21], [25].

Future studies should focus on model similarity scores and extraction resistance metrics to improve model security, providing quantitative benchmarks on a model's resistance to extraction. Query Behavior Analysis Scores can also be useful in understanding model vulnerability [3].

AI-powered defense systems can proactively identify and stop extraction efforts, enabling machine learning models to dynamically modify protection parameters in response to anomalies [3].

IX. Conclusion

This paper examines model extraction attacks, highlighting the vulnerabilities of proprietary models in cloud-based MLaaS, federated learning, and other distributed systems. It highlights the need for adaptive and hybrid attack models and the importance of ongoing research on model extraction defenses. Emerging assault techniques that bypass static safeguards require new defense mechanisms, especially adaptive, multi-layered strategies. Consistent evaluation and enhancement of defense mechanisms across various applications depend on the creation of defined criteria for evaluating model vulnerability. Developments in adaptive defenses, federated-specific security, and real-time AI-driven detection will be critical for safeguarding models in industries like healthcare, finance, and e-commerce. Future advancements in model security will prioritize collaborative and adaptive defenses, particularly in federated and decentralized learning environments.

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